

Overview of Diffusion model

Subject: Diffusion model

$$q(x_t | x_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$

1.

$$\nabla_x \ln p(x|y) = \nabla_x \ln p(x) \text{ score} + \nabla_x \ln p(y|x) \text{ classifier guidance} \quad \nabla_x \ln p(x|y) = \underbrace{\nabla_x \ln p(x)}_{\text{score}} + \underbrace{\nabla_x \ln p(y|x)}_{\text{classifier guidance}}$$

2.

Sample (x_0, z_0, c) , where x_0 is the high-resolution image, z_0 is the same image but scaled down to a low-resolution, and c is the conditioning, which can be the caption of the image, the class of the image, etc. Sample two white noises ϵ_x, ϵ_z , two time-steps t_x, t_z . Compute the noisy versions of the high-resolution and low-resolution images:
$$\begin{cases} x_{t_x} = \alpha_{t_x} x_0 + \sigma_{t_x} \epsilon_x \\ z_{t_z} = \alpha_{t_z} z_0 + \sigma_{t_z} \epsilon_z \end{cases}$$
. Train the denoising network to predict ϵ_x given $x_{t_x}, z_{t_z}, t_x, t_z, c$. That is, apply gradient descent on θ on the L2 loss
$$\|\epsilon_{\theta}(x_{t_x}, z_{t_z}, t_x, t_z, c) - \epsilon_x\|_2^2$$
.

3.

$$\nabla_x \ln p_\gamma(x|y) = (1 - \gamma) \nabla_x \ln p(x) + \gamma \nabla_x \ln p(x|y)$$

4.

Non-equilibrium thermodynamics Diffusion models were introduced in 2015 as a method to train a model that can sample from a highly complex probability distribution. They used techniques from non-equilibrium thermodynamics, especially diffusion. Consider, for example, how one might model the distribution of all naturally occurring photos. Each image is a point in the space of all images, and the distribution of naturally occurring photos is a "cloud" in space, which, by repeatedly adding noise to the images, diffuses out to the rest of the image space, until the cloud becomes all but indistinguishable from a Gaussian distribution $\mathcal{N}(0, I)$. A model that can approximately undo the diffusion can then be used to sample from the original distribution. This is studied in "non-equilibrium" thermodynamics, as the starting distribution is not in equilibrium, unlike the final distribution. The equilibrium distribution is the Gaussian distribution $\mathcal{N}(0, I)$, with pdf $p(x) \propto e^{-\frac{1}{2} x^T x}$.

$e^{-\frac{1}{2}\|x\|^2}$. This is just the Maxwell–Boltzmann distribution of particles in a potential well $V(x) = \frac{1}{2}\|x\|^2$ at temperature 1. The initial distribution, being very much out of equilibrium, would diffuse towards the equilibrium distribution, making biased random steps that are a sum of pure randomness (like a Brownian walker) and gradient descent down the potential well. The randomness is necessary: if the particles were to undergo only gradient descent, then they will all fall to the origin, collapsing the distribution.

5.

Samplers Given a diffusion model, one may regard it either as a continuous process, and sample from it by integrating a SDE, or one can regard it as a discrete process, and sample from it by iterating the discrete steps. The choice of the "noise schedule" β_t can also affect the quality of samples. A noise schedule is a function that sends a natural number to a noise level: $t \mapsto \beta_t$, $t \in \{1, 2, \dots\}$, $\beta \in (0, 1)$. A noise schedule is more often specified by a map $t \mapsto \sigma_t$. The two definitions are equivalent, since $\beta_t = 1 - 1 - \sigma_{t-1}^2 = 1 - \sigma_t^2$. In the DDPM perspective, one can use the DDPM itself (with noise), or DDIM (with adjustable amount of noise). The case where one adds noise is sometimes called ancestral sampling. One can interpolate between noise and no noise. The amount of noise is denoted η ("eta value") in the DDIM paper, with $\eta = 0$ denoting no noise (as in deterministic DDIM), and $\eta = 1$ denoting full noise (as in DDPM). In the perspective of SDE, one can use any of the numerical integration methods, such as Euler–Maruyama method, Heun's method, linear multistep methods, etc. Just as in the discrete case, one can add an adjustable amount of noise during the integration. A survey and comparison of samplers in the context of image generation is in.

6.

With temperature The classifier-guided diffusion model samples from $p(x|y)$, which is concentrated around the maximum a posteriori estimate $\arg \max_x p(x|y)$. If we want to force the model to move towards the maximum likelihood estimate $\arg \max_x p(y|x)$, we can use

7.

$\mu_t(x_t, x_0) := \alpha_t(1 - \alpha_{t-1})x_t + \alpha_{t-1}(1 - \alpha_t)x_0 + \sigma_t^2$

8.

To construct a probability path, we start by construct a conditional probability path $p_t(x|z)$ and the corresponding conditional velocity field $v_t(x|z)$ on some conditional distribution $q(z)$. A natural choice is the Gaussian conditional probability path: